Neural Networks Fail to Learn Periodic Functions and How to Fix It

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Task/Model Motivation

- Periodic functions are crucial in modeling natural and societal phenomena (e.g., biological clocks, economic cycles).
- Standard neural networks (e.g., with ReLU, tanh) struggle to extrapolate periodic functions beyond training data.
- Need for a flexible model that handles both periodic and non-periodic functions without prior knowledge of periodicity.

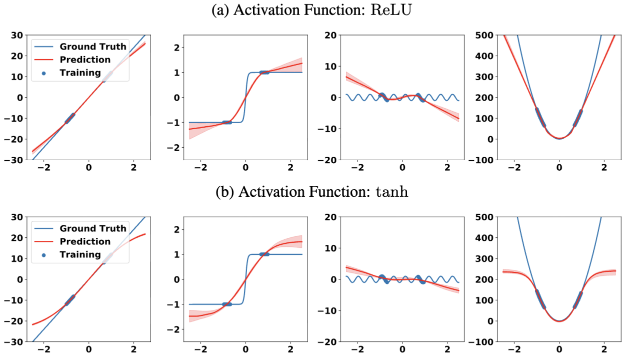

Figure: Exploration of how different activation functions extrapolate various basic function types: $y = x$ (first column), $y = \tanh(x)$ (second column), $y = \sin(x)$ (third column), and $y = x^2$ (last column). The red curves represents the median model prediction and the shaded regions show the 90% credibility interval from 21 independent runs. Note that the horizontal range is re-scaled so that the training data lies between -1 and 1.

Formal Problem Statement

Definition (Feedforward Neural Network)

Let $f_\sigma(x) = W_h \sigma \dots \sigma W_1 x$ be a function from $\mathbb{R}^{d_1} \to \mathbb{R}^{d_{h+1}}$, where σ is the activation function applied element-wise to its input vector, and $W_i \in \mathbb{R}^{d_i \times d_{i+1}}$. $f_\sigma(x)$ is called a feedforward neural network with activation function σ, and d_1 is called the input dimension, and d_{h+1} is the output dimension.

Formal Problem Statement

Theorem (ReLU Extrapolation)

Consider a feed forward network $f_{ReLU}(x)$, with arbitrary but fixed depth h and widths $d_1, \ldots, d_{h+1}$. Then

$$\lim_{z \to \infty} \| f_{ReLU}(zu) - zW_u u - b_u \| = 0,$$

where z is a real scalar, u is any unit vector of dimension d_1, and $W_u \in \mathbb{R}^{d_1 \times d_h}$ is a constant matrix only dependent on u.

Formal Problem Statement

Theorem (Tanh Extrapolation)

Consider a feed forward network $f_{tanh}(x)$, with arbitrarily fixed depth h and widths $d_1, \ldots, d_{h+1}$. Then

$$\lim_{z \to \infty} \|f_{tanh}(zu) - v_u\| = 0,$$

where z is a real scalar, u is any unit vector of dimension d_1, and $v_u \in \mathbb{R}^{d_{h+1}}$ is a constant vector that only depends on u.

Theory and Method Description

- **Theory**: Standard activations (ReLU, tanh) extrapolate linearly or to constants, unsuitable for periodic functions (Theorems 1 and 2).
- **Proposed Method**: Introduce *Snake* activation function: $\text{Snake}_a(x) = x + \frac{1}{a}\sin^2(ax)$.
- **Advantages**: Periodic inductive bias, monotonicity, and second-order non-linear term for better approximation.
- **Initialization**: Adjusted for unit variance, with learnable frequency parameter a.

Theory and Method Description

Theorem (Snake Universal Extrapolation)

Let $f(x)$ be a piecewise C^1 periodic function with period L. Then, a Snake neural network, f_{w_N}, with one hidden layer and with width N can converge to $f(x)$ uniformly as $N \to \infty$, i.e., there exists parameters w_N for all $N \in \mathbb{Z}^+$ such that

$$f(x) = \lim_{N \to \infty} f_{w_N}(x)$$

for all $x \in \mathbb{R}$, i.e., the convergence is point-wise. If $f(x)$ is continuous, then the convergence is uniform.

Initialization

Let $W \in \mathbb{R}^{d_1 \times d_2}$, whose input activations are $h \in \mathbb{R}^{d_2}$ with a unit variance for each of its element, and the goal is to set the variance of each element in W such that Snake(Wx) has a unit variance. To leading order, Snake looks like an identity function, and so one can make this approximation in finding the required variance: $\mathbb{E}\left(\sum_j^{d_2} W_{ij}x_j\right)^2 = 1$ which gives $\mathbb{E}[W_{ij}^2] = 1/d$. If we use uniform distribution to initialize W, then we should sample from $Uniform(-\sqrt{\frac{3}{d}}, \sqrt{\frac{3}{d}})$, which is a factor of $\sqrt{2}$ smaller in range than the Kaiming uniform initialization. The variance of expected value of $x + \frac{\sin^2(ax)}{a}$ under a standard normal distribution is $\sigma_a^2 = 1 + \frac{1+e^{-8a^2} - 2e^{-4a^2}}{8a^2}$, which is maximized at $a_{max} \approx 0.56045$.

Experiments, Examples, Applications

- **Regression**: Snake successfully extrapolates periodic functions (e.g., $\sin(x)$) unlike ReLU/tanh
- **Image Classification**: Competitive with ReLU on CIFAR-10 using ResNet-18/101
- **Real-World Applications**:
  - Atmospheric temperature prediction (Minamitorishima): Captures weekly cycles
  - Human body temperature: Models circadian rhythm with limited data
  - Financial data: Predicts Wilshire 5000 Index, capturing economic cycles
- **RNN Comparison**: Snake outperforms RNNs in time series tasks with less computational cost

Minamitorishima

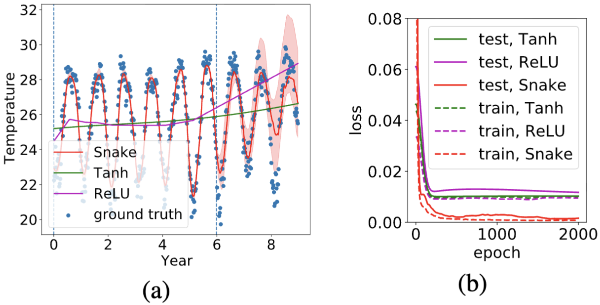

Figure: Experiment on the atmospheric data. (a) Regressing the mean weekly temperature evolution of Minamitorishima with different activation functions. For Snake, a is treated as a learnable parameter and the red contour shows the 90

Financial Data

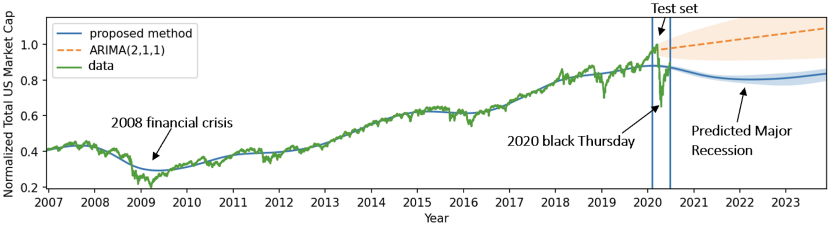

Figure: Prediction of Wilshire 5000 index, an indicator of the US and global economy.

Literature

- Hornik et al. (1989): Universal approximation theorem for neural networks.
- He et al. (2015): Kaiming initialization for deep networks.
- Ramachandran et al. (2017): Swish activation function.
- Parascandolo et al. (2016): Sine as activation function.
- Silvescu (1999), Zhumekenov et al. (2019): Fourier neural networks.
- Full references in paper: `https://proceedings.neurips.cc/paper/2020/file/1160453108d3e537255e9f7b931f4e90-Paper.pdf`